

Mathematics for Data Science & Machine Learning

COMPREHENSIVE MASTER PROBLEM BANK (QUESTIONS ONLY — CORRECTED EASY TIER)

Curriculum Focus: Foundational Course Exam Bank

Target Domains: Linear Algebra, Transformations, Metrics, Geometry, Inference, Calculus

Total Items: 350 Rigorous Problems (50 Per Main Domain)

Format: High Density Text Collection

Syllabus Overview

Domain Topic Section	Core Concepts Covered	Easy Range	Intermediate Range	Difficult Range
1. Linear Algebra: Vector Fundamentals	Scalars, Vectors, Addition, Hadamard, Dot Product, Cosine Similarity, Vector DBs	1 - 15	16 - 35	36 - 50
2. Transformations, Inverses & Eigenvalues	Linear Transformations, Projections, Unit Vectors, Matrix Inversion, Eigenvectors/Values	1 - 15	16 - 35	36 - 50
3. Coordinate Geometry of Data	Lines in R^2 , Planes in R^3 , Hyperplanes in R^d , Normal Vectors, Hyperplane Margins	1 - 15	16 - 35	36 - 50
4. Descriptive Statistics & Foundations	Central Tendency, Dispersion, Bessel's Correction, Percentiles, Covariance, Skewness	1 - 15	16 - 35	36 - 50
5. Probability Theory & Distributions	Probability Rules, PMF/PDF/CDF, Bernoulli, Binomial, Poisson, Gaussian, Z-Scores, CLT	1 - 15	16 - 35	36 - 50
6. Inferential Statistics & Hypothesis Testing	Z/T Tests, Type I/II Errors, Bayes' Theorem, Confidence Intervals, Chi-Square, ANOVA	1 - 15	16 - 35	36 - 50
7. Differential Calculus & Optimization	Slopes, Limits, Power/Product/Chain Rules, Tangents, Gradients for Machine Learning	1 - 15	16 - 35	36 - 50

Module 1: Linear Algebra - Vector & Matrix Fundamentals

- 1.1 EASY** Identify if the following values in an ML pipeline are scalars or vectors: (a) Learning Rate $\eta = 0.01$, (b) Weights $W = [0.4, -1.2, 0.85]$, (c) Loss value $L = 0.24$.
- 1.2 EASY** Given embedding vectors $u = [5, -1, 4]$ and $v = [1, 7, -2]$, calculate the standard vector addition result $w = u + v$.
- 1.3 EASY** For a feature row vector $x = [6, 4, 6]$, perform scalar multiplication with scaling coefficient $c = 0.5$.
- 1.4 EASY** Identify if the following values in an ML pipeline are scalars or vectors: (a) Learning Rate $\eta = 0.01$, (b) Weights $W = [0.4, -1.2, 0.85]$, (c) Loss value $L = 0.24$.
- 1.5 EASY** Given embedding vectors $u = [8, -1, 4]$ and $v = [1, 10, -2]$, calculate the standard vector addition result $w = u + v$.
- 1.6 EASY** For a feature row vector $x = [12, 4, 6]$, perform scalar multiplication with scaling coefficient $c = 0.5$.
- 1.7 EASY** Identify if the following values in an ML pipeline are scalars or vectors: (a) Learning Rate $\eta = 0.01$, (b) Weights $W = [0.4, -1.2, 0.85]$, (c) Loss value $L = 0.24$.
- 1.8 EASY** Given embedding vectors $u = [11, -1, 4]$ and $v = [1, 13, -2]$, calculate the standard vector addition result $w = u + v$.
- 1.9 EASY** For a feature row vector $x = [18, 4, 6]$, perform scalar multiplication with scaling coefficient $c = 0.5$.
- 1.10 EASY** Identify if the following values in an ML pipeline are scalars or vectors: (a) Learning Rate $\eta = 0.01$, (b) Weights $W = [0.4, -1.2, 0.85]$, (c) Loss value $L = 0.24$.
- 1.11 EASY** Given embedding vectors $u = [14, -1, 4]$ and $v = [1, 16, -2]$, calculate the standard vector addition result $w = u + v$.
- 1.12 EASY** For a feature row vector $x = [24, 4, 6]$, perform scalar multiplication with scaling coefficient $c = 0.5$.
- 1.13 EASY** Identify if the following values in an ML pipeline are scalars or vectors: (a) Learning Rate $\eta = 0.01$, (b) Weights $W = [0.4, -1.2, 0.85]$, (c) Loss value $L = 0.24$.
- 1.14 EASY** Given embedding vectors $u = [17, -1, 4]$ and $v = [1, 19, -2]$, calculate the standard vector addition result $w = u + v$.
- 1.15 EASY** For a feature row vector $x = [30, 4, 6]$, perform scalar multiplication with scaling coefficient $c = 0.5$.
- 1.16 INTERMEDIATE** In a neural network layer, the input vector is $x = [1, 16]$ and weight matrix is $W = \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$. Calculate the transformed layer output vector $z = Wx$.
- 1.17 INTERMEDIATE** Compute the element-wise (Hadamard) product $a \odot b$ for dropout mask vectors $a = [1, 0, 1, 1]$ and activations $b = [25.5, 4.1, -1.0, 3.2]$.

- 1.18** **INTERMEDIATE** A vector database returns text embeddings: $A = [1, 18, 2]$ and $B = [2, 0, 1]$. Calculate the exact numeric *Cosine Similarity*.
- 1.19** **INTERMEDIATE** In a neural network layer, the input vector is $x = [1, 19]$ and weight matrix is $W = \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$. Calculate the transformed layer output vector $z = Wx$.
- 1.20** **INTERMEDIATE** Compute the element-wise (Hadamard) product $a \odot b$ for dropout mask vectors $a = [1, 0, 1, 1]$ and activations $b = [30.0, 4.1, -1.0, 3.2]$.
- 1.21** **INTERMEDIATE** A vector database returns text embeddings: $A = [1, 21, 2]$ and $B = [2, 0, 1]$. Calculate the exact numeric *Cosine Similarity*.
- 1.22** **INTERMEDIATE** In a neural network layer, the input vector is $x = [1, 22]$ and weight matrix is $W = \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$. Calculate the transformed layer output vector $z = Wx$.
- 1.23** **INTERMEDIATE** Compute the element-wise (Hadamard) product $a \odot b$ for dropout mask vectors $a = [1, 0, 1, 1]$ and activations $b = [34.5, 4.1, -1.0, 3.2]$.
- 1.24** **INTERMEDIATE** A vector database returns text embeddings: $A = [1, 24, 2]$ and $B = [2, 0, 1]$. Calculate the exact numeric *Cosine Similarity*.
- 1.25** **INTERMEDIATE** In a neural network layer, the input vector is $x = [1, 25]$ and weight matrix is $W = \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$. Calculate the transformed layer output vector $z = Wx$.
- 1.26** **INTERMEDIATE** Compute the element-wise (Hadamard) product $a \odot b$ for dropout mask vectors $a = [1, 0, 1, 1]$ and activations $b = [39.0, 4.1, -1.0, 3.2]$.
- 1.27** **INTERMEDIATE** A vector database returns text embeddings: $A = [1, 27, 2]$ and $B = [2, 0, 1]$. Calculate the exact numeric *Cosine Similarity*.
- 1.28** **INTERMEDIATE** In a neural network layer, the input vector is $x = [1, 28]$ and weight matrix is $W = \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$. Calculate the transformed layer output vector $z = Wx$.
- 1.29** **INTERMEDIATE** Compute the element-wise (Hadamard) product $a \odot b$ for dropout mask vectors $a = [1, 0, 1, 1]$ and activations $b = [43.5, 4.1, -1.0, 3.2]$.
- 1.30** **INTERMEDIATE** A vector database returns text embeddings: $A = [1, 30, 2]$ and $B = [2, 0, 1]$. Calculate the exact numeric *Cosine Similarity*.
- 1.31** **INTERMEDIATE** In a neural network layer, the input vector is $x = [1, 31]$ and weight matrix is $W = \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$. Calculate the transformed layer output vector $z = Wx$.
- 1.32** **INTERMEDIATE** Compute the element-wise (Hadamard) product $a \odot b$ for dropout mask vectors $a = [1, 0, 1, 1]$ and activations $b = [48.0, 4.1, -1.0, 3.2]$.
- 1.33** **INTERMEDIATE** A vector database returns text embeddings: $A = [1, 33, 2]$ and $B = [2, 0, 1]$. Calculate the exact numeric *Cosine Similarity*.
- 1.34** **INTERMEDIATE** In a neural network layer, the input vector is $x = [1, 34]$ and weight matrix is $W = \begin{bmatrix} 3 & 5 \\ 1 & -2 \end{bmatrix}$. Calculate the transformed layer output vector $z = Wx$.

- 1.35** **INTERMEDIATE** Compute the element-wise (Hadamard) product $\mathbf{a} \odot \mathbf{b}$ for dropout mask vectors $\mathbf{a} = [1, 0, 1, 1]$ and activations $\mathbf{b} = [52.5, 4.1, -1.0, 3.2]$.
- 1.36** **DIFFICULT** Prove that if two user preference vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, their cosine similarity value evaluates exactly to 0. Explain how this filters recommendation outputs.
- 1.37** **DIFFICULT** Given a batch matrix $\mathbf{X}$ of shape $(7, 2)$ and weights $\mathbf{W}$ of shape $(2, 4)$, calculate the product $\mathbf{Z} = \mathbf{XW}$. Prove that the transpose identity holds true.
- 1.38** **DIFFICULT** Prove that if two user preference vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, their cosine similarity value evaluates exactly to 0. Explain how this filters recommendation outputs.
- 1.39** **DIFFICULT** Given a batch matrix $\mathbf{X}$ of shape $(9, 2)$ and weights $\mathbf{W}$ of shape $(2, 4)$, calculate the product $\mathbf{Z} = \mathbf{XW}$. Prove that the transpose identity holds true.
- 1.40** **DIFFICULT** Prove that if two user preference vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, their cosine similarity value evaluates exactly to 0. Explain how this filters recommendation outputs.
- 1.41** **DIFFICULT** Given a batch matrix $\mathbf{X}$ of shape $(11, 2)$ and weights $\mathbf{W}$ of shape $(2, 4)$, calculate the product $\mathbf{Z} = \mathbf{XW}$. Prove that the transpose identity holds true.
- 1.42** **DIFFICULT** Prove that if two user preference vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, their cosine similarity value evaluates exactly to 0. Explain how this filters recommendation outputs.
- 1.43** **DIFFICULT** Given a batch matrix $\mathbf{X}$ of shape $(13, 2)$ and weights $\mathbf{W}$ of shape $(2, 4)$, calculate the product $\mathbf{Z} = \mathbf{XW}$. Prove that the transpose identity holds true.
- 1.44** **DIFFICULT** Prove that if two user preference vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, their cosine similarity value evaluates exactly to 0. Explain how this filters recommendation outputs.
- 1.45** **DIFFICULT** Given a batch matrix $\mathbf{X}$ of shape $(15, 2)$ and weights $\mathbf{W}$ of shape $(2, 4)$, calculate the product $\mathbf{Z} = \mathbf{XW}$. Prove that the transpose identity holds true.
- 1.46** **DIFFICULT** Prove that if two user preference vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, their cosine similarity value evaluates exactly to 0. Explain how this filters recommendation outputs.
- 1.47** **DIFFICULT** Given a batch matrix $\mathbf{X}$ of shape $(17, 2)$ and weights $\mathbf{W}$ of shape $(2, 4)$, calculate the product $\mathbf{Z} = \mathbf{XW}$. Prove that the transpose identity holds true.
- 1.48** **DIFFICULT** Prove that if two user preference vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, their cosine similarity value evaluates exactly to 0. Explain how this filters recommendation outputs.
- 1.49** **DIFFICULT** Given a batch matrix $\mathbf{X}$ of shape $(19, 2)$ and weights $\mathbf{W}$ of shape $(2, 4)$, calculate the product $\mathbf{Z} = \mathbf{XW}$. Prove that the transpose identity holds true.
- 1.50** **DIFFICULT** Prove that if two user preference vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, their cosine similarity value evaluates exactly to 0. Explain how this filters recommendation outputs.

Module 2: Linear Transformations, Matrix Inverses & Eigenvalues

- 2.1 EASY** Normalize the raw feature vector $\mathbf{x} = [1, 3]$ into a standard unit vector $\hat{\mathbf{u}}$ with an L2 norm equal to 1.
- 2.2 EASY** Find the vector length (L2 Norm) $\|\mathbf{v}\|$ for the 2D error vector component $\mathbf{v} = [-2, 4]$.
- 2.3 EASY** Normalize the raw feature vector $\mathbf{x} = [3, 3]$ into a standard unit vector $\hat{\mathbf{u}}$ with an L2 norm equal to 1.
- 2.4 EASY** Find the vector length (L2 Norm) $\|\mathbf{v}\|$ for the 2D error vector component $\mathbf{v} = [-4, 4]$.
- 2.5 EASY** Normalize the raw feature vector $\mathbf{x} = [5, 3]$ into a standard unit vector $\hat{\mathbf{u}}$ with an L2 norm equal to 1.
- 2.6 EASY** Find the vector length (L2 Norm) $\|\mathbf{v}\|$ for the 2D error vector component $\mathbf{v} = [-6, 4]$.
- 2.7 EASY** Normalize the raw feature vector $\mathbf{x} = [7, 3]$ into a standard unit vector $\hat{\mathbf{u}}$ with an L2 norm equal to 1.
- 2.8 EASY** Find the vector length (L2 Norm) $\|\mathbf{v}\|$ for the 2D error vector component $\mathbf{v} = [-8, 4]$.
- 2.9 EASY** Normalize the raw feature vector $\mathbf{x} = [9, 3]$ into a standard unit vector $\hat{\mathbf{u}}$ with an L2 norm equal to 1.
- 2.10 EASY** Find the vector length (L2 Norm) $\|\mathbf{v}\|$ for the 2D error vector component $\mathbf{v} = [-10, 4]$.
- 2.11 EASY** Normalize the raw feature vector $\mathbf{x} = [11, 3]$ into a standard unit vector $\hat{\mathbf{u}}$ with an L2 norm equal to 1.
- 2.12 EASY** Find the vector length (L2 Norm) $\|\mathbf{v}\|$ for the 2D error vector component $\mathbf{v} = [-12, 4]$.
- 2.13 EASY** Normalize the raw feature vector $\mathbf{x} = [13, 3]$ into a standard unit vector $\hat{\mathbf{u}}$ with an L2 norm equal to 1.
- 2.14 EASY** Find the vector length (L2 Norm) $\|\mathbf{v}\|$ for the 2D error vector component $\mathbf{v} = [-14, 4]$.
- 2.15 EASY** Normalize the raw feature vector $\mathbf{x} = [15, 3]$ into a standard unit vector $\hat{\mathbf{u}}$ with an L2 norm equal to 1.
- 2.16 INTERMEDIATE** Verify whether the mapping transformation function $T(\mathbf{x}, \mathbf{y}) = (16\mathbf{x} + \mathbf{y}, \mathbf{x} - 3\mathbf{y})$ satisfies criteria for a linear transformation.
- 2.17 INTERMEDIATE** Calculate the orthogonal projection vector $\text{proj}_{\mathbf{v}}(\mathbf{u})$ of a data point vector $\mathbf{u} = [17, 3]$ onto a unit direction vector axis $\mathbf{v} = [1, 0]$.
- 2.18 INTERMEDIATE** Given a 2x2 matrix transformation mapping $\mathbf{A} = \begin{bmatrix} 8 & 1 \\ 1 & 3 \end{bmatrix}$, calculate its explicit matrix inverse $\mathbf{A}^{-1}$ dynamically.
- 2.19 INTERMEDIATE** Verify whether the mapping transformation function $T(\mathbf{x}, \mathbf{y}) = (19\mathbf{x} + \mathbf{y}, \mathbf{x} - 3\mathbf{y})$ satisfies criteria for a linear transformation.
- 2.20 INTERMEDIATE** Calculate the orthogonal projection vector $\text{proj}_{\mathbf{v}}(\mathbf{u})$ of a data point vector $\mathbf{u} = [20, 3]$ onto a unit direction vector axis $\mathbf{v} = [1, 0]$.

- 2.21** **INTERMEDIATE** Given a 2x2 matrix transformation mapping $A = \begin{bmatrix} 11 & 1 \\ 1 & 3 \end{bmatrix}$, calculate its explicit matrix inverse A^{-1} dynamically.
- 2.22** **INTERMEDIATE** Verify whether the mapping transformation function $T(x, y) = (22x + y, x - 3y)$ satisfies criteria for a linear transformation.
- 2.23** **INTERMEDIATE** Calculate the orthogonal projection vector $\text{proj}_{\mathbf{v}}(\mathbf{u})$ of a data point vector $\mathbf{u} = [23, 3]$ onto a unit direction vector axis $\mathbf{v} = [1, 0]$.
- 2.24** **INTERMEDIATE** Given a 2x2 matrix transformation mapping $A = \begin{bmatrix} 14 & 1 \\ 1 & 3 \end{bmatrix}$, calculate its explicit matrix inverse A^{-1} dynamically.
- 2.25** **INTERMEDIATE** Verify whether the mapping transformation function $T(x, y) = (25x + y, x - 3y)$ satisfies criteria for a linear transformation.
- 2.26** **INTERMEDIATE** Calculate the orthogonal projection vector $\text{proj}_{\mathbf{v}}(\mathbf{u})$ of a data point vector $\mathbf{u} = [26, 3]$ onto a unit direction vector axis $\mathbf{v} = [1, 0]$.
- 2.27** **INTERMEDIATE** Given a 2x2 matrix transformation mapping $A = \begin{bmatrix} 17 & 1 \\ 1 & 3 \end{bmatrix}$, calculate its explicit matrix inverse A^{-1} dynamically.
- 2.28** **INTERMEDIATE** Verify whether the mapping transformation function $T(x, y) = (28x + y, x - 3y)$ satisfies criteria for a linear transformation.
- 2.29** **INTERMEDIATE** Calculate the orthogonal projection vector $\text{proj}_{\mathbf{v}}(\mathbf{u})$ of a data point vector $\mathbf{u} = [29, 3]$ onto a unit direction vector axis $\mathbf{v} = [1, 0]$.
- 2.30** **INTERMEDIATE** Given a 2x2 matrix transformation mapping $A = \begin{bmatrix} 20 & 1 \\ 1 & 3 \end{bmatrix}$, calculate its explicit matrix inverse A^{-1} dynamically.
- 2.31** **INTERMEDIATE** Verify whether the mapping transformation function $T(x, y) = (31x + y, x - 3y)$ satisfies criteria for a linear transformation.
- 2.32** **INTERMEDIATE** Calculate the orthogonal projection vector $\text{proj}_{\mathbf{v}}(\mathbf{u})$ of a data point vector $\mathbf{u} = [32, 3]$ onto a unit direction vector axis $\mathbf{v} = [1, 0]$.
- 2.33** **INTERMEDIATE** Given a 2x2 matrix transformation mapping $A = \begin{bmatrix} 23 & 1 \\ 1 & 3 \end{bmatrix}$, calculate its explicit matrix inverse A^{-1} dynamically.
- 2.34** **INTERMEDIATE** Verify whether the mapping transformation function $T(x, y) = (34x + y, x - 3y)$ satisfies criteria for a linear transformation.
- 2.35** **INTERMEDIATE** Calculate the orthogonal projection vector $\text{proj}_{\mathbf{v}}(\mathbf{u})$ of a data point vector $\mathbf{u} = [35, 3]$ onto a unit direction vector axis $\mathbf{v} = [1, 0]$.
- 2.36** **DIFFICULT** Find the characteristic **Eigenvalues** (λ) for the sample data covariance matrix structure $\Sigma = \begin{bmatrix} 6 & 2 \\ 2 & 7 \end{bmatrix}$.
- 2.37** **DIFFICULT** For the matrix $M = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, find the corresponding normalized **Eigenvectors** for each calculated eigenvalue.

2.38 **DIFFICULT** Find the characteristic *Eigenvalues* (λ) for the sample data covariance matrix structure $\Sigma = \begin{bmatrix} 8 & 2 \\ 2 & 7 \end{bmatrix}$.

2.39 **DIFFICULT** For the matrix $M = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$, find the corresponding normalized *Eigenvectors* for each calculated eigenvalue.

2.40 **DIFFICULT** Find the characteristic *Eigenvalues* (λ) for the sample data covariance matrix structure $\Sigma = \begin{bmatrix} 10 & 2 \\ 2 & 7 \end{bmatrix}$.

2.41 **DIFFICULT** For the matrix $M = \begin{bmatrix} 6 & 1 \\ 1 & 2 \end{bmatrix}$, find the corresponding normalized *Eigenvectors* for each calculated eigenvalue.

2.42 **DIFFICULT** Find the characteristic *Eigenvalues* (λ) for the sample data covariance matrix structure $\Sigma = \begin{bmatrix} 12 & 2 \\ 2 & 7 \end{bmatrix}$.

2.43 **DIFFICULT** For the matrix $M = \begin{bmatrix} 8 & 1 \\ 1 & 2 \end{bmatrix}$, find the corresponding normalized *Eigenvectors* for each calculated eigenvalue.

2.44 **DIFFICULT** Find the characteristic *Eigenvalues* (λ) for the sample data covariance matrix structure $\Sigma = \begin{bmatrix} 14 & 2 \\ 2 & 7 \end{bmatrix}$.

2.45 **DIFFICULT** For the matrix $M = \begin{bmatrix} 10 & 1 \\ 1 & 2 \end{bmatrix}$, find the corresponding normalized *Eigenvectors* for each calculated eigenvalue.

2.46 **DIFFICULT** Find the characteristic *Eigenvalues* (λ) for the sample data covariance matrix structure $\Sigma = \begin{bmatrix} 16 & 2 \\ 2 & 7 \end{bmatrix}$.

2.47 **DIFFICULT** For the matrix $M = \begin{bmatrix} 12 & 1 \\ 1 & 2 \end{bmatrix}$, find the corresponding normalized *Eigenvectors* for each calculated eigenvalue.

2.48 **DIFFICULT** Find the characteristic *Eigenvalues* (λ) for the sample data covariance matrix structure $\Sigma = \begin{bmatrix} 18 & 2 \\ 2 & 7 \end{bmatrix}$.

2.49 **DIFFICULT** For the matrix $M = \begin{bmatrix} 14 & 1 \\ 1 & 2 \end{bmatrix}$, find the corresponding normalized *Eigenvectors* for each calculated eigenvalue.

2.50 **DIFFICULT** Find the characteristic *Eigenvalues* (λ) for the sample data covariance matrix structure $\Sigma = \begin{bmatrix} 20 & 2 \\ 2 & 7 \end{bmatrix}$.

Module 3: Coordinate Geometry of Data - Lines, Planes & Hyperplanes

3.1 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $1x - 4y + 12 = 0$.

3.2 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $2x - 4y + 12 = 0$.

3.3 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $3x - 4y + 12 = 0$.

3.4 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $4x - 4y + 12 = 0$.

3.5 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $5x - 4y + 12 = 0$.

3.6 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $6x - 4y + 12 = 0$.

3.7 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $7x - 4y + 12 = 0$.

3.8 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $8x - 4y + 12 = 0$.

3.9 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $9x - 4y + 12 = 0$.

3.10 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $10x - 4y + 12 = 0$.

3.11 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $11x - 4y + 12 = 0$.

3.12 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $12x - 4y + 12 = 0$.

3.13 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $13x - 4y + 12 = 0$.

3.14 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $14x - 4y + 12 = 0$.

3.15 **EASY** Determine the structural slope parameter and coordinate y-intercept for the linear regression line described by $15x - 4y + 12 = 0$.

3.16 **INTERMEDIATE** Construct the scalar algebraic equation for a 3D decision plane passing through data point $P(1, -2, 3)$ with normal vector $n = [4, 2, -1]$.

- 3.17** **INTERMEDIATE** Calculate the spatial shortest distance from an outlier data point at position $(17, 6)$ to the regression boundary line given by $y = 2x - 1$.
- 3.18** **INTERMEDIATE** Construct the scalar algebraic equation for a 3D decision plane passing through data point $P(3, -2, 3)$ with normal vector $\mathbf{n} = [4, 2, -1]$.
- 3.19** **INTERMEDIATE** Calculate the spatial shortest distance from an outlier data point at position $(19, 6)$ to the regression boundary line given by $y = 2x - 1$.
- 3.20** **INTERMEDIATE** Construct the scalar algebraic equation for a 3D decision plane passing through data point $P(5, -2, 3)$ with normal vector $\mathbf{n} = [4, 2, -1]$.
- 3.21** **INTERMEDIATE** Calculate the spatial shortest distance from an outlier data point at position $(21, 6)$ to the regression boundary line given by $y = 2x - 1$.
- 3.22** **INTERMEDIATE** Construct the scalar algebraic equation for a 3D decision plane passing through data point $P(7, -2, 3)$ with normal vector $\mathbf{n} = [4, 2, -1]$.
- 3.23** **INTERMEDIATE** Calculate the spatial shortest distance from an outlier data point at position $(23, 6)$ to the regression boundary line given by $y = 2x - 1$.
- 3.24** **INTERMEDIATE** Construct the scalar algebraic equation for a 3D decision plane passing through data point $P(9, -2, 3)$ with normal vector $\mathbf{n} = [4, 2, -1]$.
- 3.25** **INTERMEDIATE** Calculate the spatial shortest distance from an outlier data point at position $(25, 6)$ to the regression boundary line given by $y = 2x - 1$.
- 3.26** **INTERMEDIATE** Construct the scalar algebraic equation for a 3D decision plane passing through data point $P(11, -2, 3)$ with normal vector $\mathbf{n} = [4, 2, -1]$.
- 3.27** **INTERMEDIATE** Calculate the spatial shortest distance from an outlier data point at position $(27, 6)$ to the regression boundary line given by $y = 2x - 1$.
- 3.28** **INTERMEDIATE** Construct the scalar algebraic equation for a 3D decision plane passing through data point $P(13, -2, 3)$ with normal vector $\mathbf{n} = [4, 2, -1]$.
- 3.29** **INTERMEDIATE** Calculate the spatial shortest distance from an outlier data point at position $(29, 6)$ to the regression boundary line given by $y = 2x - 1$.
- 3.30** **INTERMEDIATE** Construct the scalar algebraic equation for a 3D decision plane passing through data point $P(15, -2, 3)$ with normal vector $\mathbf{n} = [4, 2, -1]$.
- 3.31** **INTERMEDIATE** Calculate the spatial shortest distance from an outlier data point at position $(31, 6)$ to the regression boundary line given by $y = 2x - 1$.
- 3.32** **INTERMEDIATE** Construct the scalar algebraic equation for a 3D decision plane passing through data point $P(17, -2, 3)$ with normal vector $\mathbf{n} = [4, 2, -1]$.
- 3.33** **INTERMEDIATE** Calculate the spatial shortest distance from an outlier data point at position $(33, 6)$ to the regression boundary line given by $y = 2x - 1$.

- 3.34** **INTERMEDIATE** Construct the scalar algebraic equation for a 3D decision plane passing through data point $P(19, -2, 3)$ with normal vector $\mathbf{n} = [4, 2, -1]$.
- 3.35** **INTERMEDIATE** Calculate the spatial shortest distance from an outlier data point at position $(35, 6)$ to the regression boundary line given by $y = 2x - 1$.
- 3.36** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^6$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.37** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^7$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.38** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^8$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.39** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^9$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.40** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^{10}$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.41** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^{11}$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.42** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^{12}$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.43** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^{13}$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.44** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^{14}$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.45** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^{15}$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.46** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^{16}$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.47** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^{17}$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.48** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^{18}$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.49** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^{19}$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.
- 3.50** **DIFFICULT** In a linear Support Vector Machine (SVM) setup, a separating hyperplane in multi-dimensional space $\mathbb{R}^{20}$ is defined by $\mathbf{w}^T \mathbf{x} + \mathbf{b} = 0$. Formulate the geometric margin distance optimizer expression.

Module 4: Descriptive Statistics & Data Foundations

- 4.1 EASY** For the sample array slice of model predictions $X = [1, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.2 EASY** For the sample array slice of model predictions $X = [2, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.3 EASY** For the sample array slice of model predictions $X = [3, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.4 EASY** For the sample array slice of model predictions $X = [4, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.5 EASY** For the sample array slice of model predictions $X = [5, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.6 EASY** For the sample array slice of model predictions $X = [6, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.7 EASY** For the sample array slice of model predictions $X = [7, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.8 EASY** For the sample array slice of model predictions $X = [8, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.9 EASY** For the sample array slice of model predictions $X = [9, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.10 EASY** For the sample array slice of model predictions $X = [10, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.11 EASY** For the sample array slice of model predictions $X = [11, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.12 EASY** For the sample array slice of model predictions $X = [12, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.13 EASY** For the sample array slice of model predictions $X = [13, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.14 EASY** For the sample array slice of model predictions $X = [14, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.15 EASY** For the sample array slice of model predictions $X = [15, 15, 12, 18, 22, 15, 30]$, compute the mean, median, and mode values.
- 4.16 INTERMEDIATE** Given a sample batch of test errors $[16, 8, 6, 10, 12]$, calculate the sample variance. Clearly explain the importance of Bessel's Correction ($n-1$).

- 4.17** **INTERMEDIATE** Extract the 5-number summary (Min, Q1, Median, Q3, Max) for the network latency distribution list: *[17, 22, 14, 15, 18, 45, 32, 28, 21]*.
- 4.18** **INTERMEDIATE** A target feature field has a mean value of 58 and median value of 42. Characterize the mathematical skewness trend.
- 4.19** **INTERMEDIATE** Given a sample batch of test errors *[19, 8, 6, 10, 12]*, calculate the sample variance. Clearly explain the importance of Bessel's Correction ($n-1$).
- 4.20** **INTERMEDIATE** Extract the 5-number summary (Min, Q1, Median, Q3, Max) for the network latency distribution list: *[20, 22, 14, 15, 18, 45, 32, 28, 21]*.
- 4.21** **INTERMEDIATE** A target feature field has a mean value of 61 and median value of 42. Characterize the mathematical skewness trend.
- 4.22** **INTERMEDIATE** Given a sample batch of test errors *[22, 8, 6, 10, 12]*, calculate the sample variance. Clearly explain the importance of Bessel's Correction ($n-1$).
- 4.23** **INTERMEDIATE** Extract the 5-number summary (Min, Q1, Median, Q3, Max) for the network latency distribution list: *[23, 22, 14, 15, 18, 45, 32, 28, 21]*.
- 4.24** **INTERMEDIATE** A target feature field has a mean value of 64 and median value of 42. Characterize the mathematical skewness trend.
- 4.25** **INTERMEDIATE** Given a sample batch of test errors *[25, 8, 6, 10, 12]*, calculate the sample variance. Clearly explain the importance of Bessel's Correction ($n-1$).
- 4.26** **INTERMEDIATE** Extract the 5-number summary (Min, Q1, Median, Q3, Max) for the network latency distribution list: *[26, 22, 14, 15, 18, 45, 32, 28, 21]*.
- 4.27** **INTERMEDIATE** A target feature field has a mean value of 67 and median value of 42. Characterize the mathematical skewness trend.
- 4.28** **INTERMEDIATE** Given a sample batch of test errors *[28, 8, 6, 10, 12]*, calculate the sample variance. Clearly explain the importance of Bessel's Correction ($n-1$).
- 4.29** **INTERMEDIATE** Extract the 5-number summary (Min, Q1, Median, Q3, Max) for the network latency distribution list: *[29, 22, 14, 15, 18, 45, 32, 28, 21]*.
- 4.30** **INTERMEDIATE** A target feature field has a mean value of 70 and median value of 42. Characterize the mathematical skewness trend.
- 4.31** **INTERMEDIATE** Given a sample batch of test errors *[31, 8, 6, 10, 12]*, calculate the sample variance. Clearly explain the importance of Bessel's Correction ($n-1$).
- 4.32** **INTERMEDIATE** Extract the 5-number summary (Min, Q1, Median, Q3, Max) for the network latency distribution list: *[32, 22, 14, 15, 18, 45, 32, 28, 21]*.
- 4.33** **INTERMEDIATE** A target feature field has a mean value of 73 and median value of 42. Characterize the mathematical skewness trend.

- 4.34** **INTERMEDIATE** Given a sample batch of test errors $[34, 8, 6, 10, 12]$, calculate the sample variance. Clearly explain the importance of Bessel's Correction ($n-1$).
- 4.35** **INTERMEDIATE** Extract the 5-number summary (Min, Q1, Median, Q3, Max) for the network latency distribution list: $[35, 22, 14, 15, 18, 45, 32, 28, 21]$.
- 4.36** **DIFFICULT** Given feature measurements $X = [1, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.37** **DIFFICULT** Given feature measurements $X = [2, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.38** **DIFFICULT** Given feature measurements $X = [3, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.39** **DIFFICULT** Given feature measurements $X = [4, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.40** **DIFFICULT** Given feature measurements $X = [5, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.41** **DIFFICULT** Given feature measurements $X = [6, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.42** **DIFFICULT** Given feature measurements $X = [7, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.43** **DIFFICULT** Given feature measurements $X = [8, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.44** **DIFFICULT** Given feature measurements $X = [9, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.45** **DIFFICULT** Given feature measurements $X = [10, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.46** **DIFFICULT** Given feature measurements $X = [11, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.47** **DIFFICULT** Given feature measurements $X = [12, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.48** **DIFFICULT** Given feature measurements $X = [13, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.49** **DIFFICULT** Given feature measurements $X = [14, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).
- 4.50** **DIFFICULT** Given feature measurements $X = [15, 3, 5]$ and $Y = [2, 6, 7]$, construct the covariance element and evaluate the normalized **Pearson Coefficient** (r).

Module 5: Probability Theory & Statistical Distributions

- 5.1** **EASY** Contrast the baseline formal properties of a discrete Probability Mass Function (PMF) against a continuous Probability Density Function (PDF) across **1** samples.
- 5.2** **EASY** An anomaly pipeline checks logs. If Event A has probability $P(A) = 0.02$ and independent Event B has $P(B) = 0.10$, evaluate joint risk $P(A \cap B)$.
- 5.3** **EASY** Contrast the baseline formal properties of a discrete Probability Mass Function (PMF) against a continuous Probability Density Function (PDF) across **3** samples.
- 5.4** **EASY** An anomaly pipeline checks logs. If Event A has probability $P(A) = 0.04$ and independent Event B has $P(B) = 0.10$, evaluate joint risk $P(A \cap B)$.
- 5.5** **EASY** Contrast the baseline formal properties of a discrete Probability Mass Function (PMF) against a continuous Probability Density Function (PDF) across **5** samples.
- 5.6** **EASY** An anomaly pipeline checks logs. If Event A has probability $P(A) = 0.06$ and independent Event B has $P(B) = 0.10$, evaluate joint risk $P(A \cap B)$.
- 5.7** **EASY** Contrast the baseline formal properties of a discrete Probability Mass Function (PMF) against a continuous Probability Density Function (PDF) across **7** samples.
- 5.8** **EASY** An anomaly pipeline checks logs. If Event A has probability $P(A) = 0.08$ and independent Event B has $P(B) = 0.10$, evaluate joint risk $P(A \cap B)$.
- 5.9** **EASY** Contrast the baseline formal properties of a discrete Probability Mass Function (PMF) against a continuous Probability Density Function (PDF) across **9** samples.
- 5.10** **EASY** An anomaly pipeline checks logs. If Event A has probability $P(A) = 0.010$ and independent Event B has $P(B) = 0.10$, evaluate joint risk $P(A \cap B)$.
- 5.11** **EASY** Contrast the baseline formal properties of a discrete Probability Mass Function (PMF) against a continuous Probability Density Function (PDF) across **11** samples.
- 5.12** **EASY** An anomaly pipeline checks logs. If Event A has probability $P(A) = 0.012$ and independent Event B has $P(B) = 0.10$, evaluate joint risk $P(A \cap B)$.
- 5.13** **EASY** Contrast the baseline formal properties of a discrete Probability Mass Function (PMF) against a continuous Probability Density Function (PDF) across **13** samples.
- 5.14** **EASY** An anomaly pipeline checks logs. If Event A has probability $P(A) = 0.014$ and independent Event B has $P(B) = 0.10$, evaluate joint risk $P(A \cap B)$.
- 5.15** **EASY** Contrast the baseline formal properties of a discrete Probability Mass Function (PMF) against a continuous Probability Density Function (PDF) across **15** samples.
- 5.16** **INTERMEDIATE** A model registers a click conversion rate of $p = 0.06$. Using the **Binomial Distribution** formula, calculate the likelihood of getting exactly 3 clicks from 20 trials.

- 5.17** **INTERMEDIATE** A server cluster gets traffic following a **Poisson Distribution** with rate $\lambda = 3$ requests/sec. Find the probability of receiving exactly 2 requests.
- 5.18** **INTERMEDIATE** Transform an raw input value of $x = 88$ into an explicit standard **Z-score** given standard parameters $\mu = 70$ and $\sigma = 10$.
- 5.19** **INTERMEDIATE** A model registers a click conversion rate of $p = 0.09$. Using the **Binomial Distribution** formula, calculate the likelihood of getting exactly 3 clicks from 20 trials.
- 5.20** **INTERMEDIATE** A server cluster gets traffic following a **Poisson Distribution** with rate $\lambda = 1$ requests/sec. Find the probability of receiving exactly 2 requests.
- 5.21** **INTERMEDIATE** Transform an raw input value of $x = 91$ into an explicit standard **Z-score** given standard parameters $\mu = 70$ and $\sigma = 10$.
- 5.22** **INTERMEDIATE** A model registers a click conversion rate of $p = 0.02$. Using the **Binomial Distribution** formula, calculate the likelihood of getting exactly 3 clicks from 20 trials.
- 5.23** **INTERMEDIATE** A server cluster gets traffic following a **Poisson Distribution** with rate $\lambda = 4$ requests/sec. Find the probability of receiving exactly 2 requests.
- 5.24** **INTERMEDIATE** Transform an raw input value of $x = 94$ into an explicit standard **Z-score** given standard parameters $\mu = 70$ and $\sigma = 10$.
- 5.25** **INTERMEDIATE** A model registers a click conversion rate of $p = 0.05$. Using the **Binomial Distribution** formula, calculate the likelihood of getting exactly 3 clicks from 20 trials.
- 5.26** **INTERMEDIATE** A server cluster gets traffic following a **Poisson Distribution** with rate $\lambda = 2$ requests/sec. Find the probability of receiving exactly 2 requests.
- 5.27** **INTERMEDIATE** Transform an raw input value of $x = 97$ into an explicit standard **Z-score** given standard parameters $\mu = 70$ and $\sigma = 10$.
- 5.28** **INTERMEDIATE** A model registers a click conversion rate of $p = 0.08$. Using the **Binomial Distribution** formula, calculate the likelihood of getting exactly 3 clicks from 20 trials.
- 5.29** **INTERMEDIATE** A server cluster gets traffic following a **Poisson Distribution** with rate $\lambda = 5$ requests/sec. Find the probability of receiving exactly 2 requests.
- 5.30** **INTERMEDIATE** Transform an raw input value of $x = 100$ into an explicit standard **Z-score** given standard parameters $\mu = 70$ and $\sigma = 10$.
- 5.31** **INTERMEDIATE** A model registers a click conversion rate of $p = 0.01$. Using the **Binomial Distribution** formula, calculate the likelihood of getting exactly 3 clicks from 20 trials.
- 5.32** **INTERMEDIATE** A server cluster gets traffic following a **Poisson Distribution** with rate $\lambda = 3$ requests/sec. Find the probability of receiving exactly 2 requests.
- 5.33** **INTERMEDIATE** Transform an raw input value of $x = 103$ into an explicit standard **Z-score** given standard parameters $\mu = 70$ and $\sigma = 10$.

- 5.34** **INTERMEDIATE** A model registers a click conversion rate of $p = 0.04$. Using the **Binomial Distribution** formula, calculate the likelihood of getting exactly 3 clicks from 20 trials.
- 5.35** **INTERMEDIATE** A server cluster gets traffic following a **Poisson Distribution** with rate $\lambda = 1$ requests/sec. Find the probability of receiving exactly 2 requests.
- 5.36** **DIFFICULT** Explain the behavioral impact of the **Central Limit Theorem (CLT)** when sampling highly non-normal data records with sample size $N = 360$.
- 5.37** **DIFFICULT** A targets column exhibits a Log-Normal shape with scaling parameter $\mu = 3.7$. Formulate the natural log transformation mapping to satisfy Gaussian requirements.
- 5.38** **DIFFICULT** Explain the behavioral impact of the **Central Limit Theorem (CLT)** when sampling highly non-normal data records with sample size $N = 380$.
- 5.39** **DIFFICULT** A targets column exhibits a Log-Normal shape with scaling parameter $\mu = 3.9$. Formulate the natural log transformation mapping to satisfy Gaussian requirements.
- 5.40** **DIFFICULT** Explain the behavioral impact of the **Central Limit Theorem (CLT)** when sampling highly non-normal data records with sample size $N = 400$.
- 5.41** **DIFFICULT** A targets column exhibits a Log-Normal shape with scaling parameter $\mu = 4.1$. Formulate the natural log transformation mapping to satisfy Gaussian requirements.
- 5.42** **DIFFICULT** Explain the behavioral impact of the **Central Limit Theorem (CLT)** when sampling highly non-normal data records with sample size $N = 420$.
- 5.43** **DIFFICULT** A targets column exhibits a Log-Normal shape with scaling parameter $\mu = 4.3$. Formulate the natural log transformation mapping to satisfy Gaussian requirements.
- 5.44** **DIFFICULT** Explain the behavioral impact of the **Central Limit Theorem (CLT)** when sampling highly non-normal data records with sample size $N = 440$.
- 5.45** **DIFFICULT** A targets column exhibits a Log-Normal shape with scaling parameter $\mu = 4.5$. Formulate the natural log transformation mapping to satisfy Gaussian requirements.
- 5.46** **DIFFICULT** Explain the behavioral impact of the **Central Limit Theorem (CLT)** when sampling highly non-normal data records with sample size $N = 460$.
- 5.47** **DIFFICULT** A targets column exhibits a Log-Normal shape with scaling parameter $\mu = 4.7$. Formulate the natural log transformation mapping to satisfy Gaussian requirements.
- 5.48** **DIFFICULT** Explain the behavioral impact of the **Central Limit Theorem (CLT)** when sampling highly non-normal data records with sample size $N = 480$.
- 5.49** **DIFFICULT** A targets column exhibits a Log-Normal shape with scaling parameter $\mu = 4.9$. Formulate the natural log transformation mapping to satisfy Gaussian requirements.
- 5.50** **DIFFICULT** Explain the behavioral impact of the **Central Limit Theorem (CLT)** when sampling highly non-normal data records with sample size $N = 500$.

Module 6: Inferential Statistics & Hypothesis Testing

- 6.1 EASY** Formulate clear mathematical statements for the Null Hypothesis (H_0) and Alternative Hypothesis (H_1) for an A/B test designed to see if a new deep learning algorithm variant alters base click conversion rates.
- 6.2 EASY** Explain the practical conceptual difference between a one-tailed and a two-tailed hypothesis test region as it applies to evaluating model variance updates.
- 6.3 EASY** Given a computed experimental p-value of **0.032**, state explicitly whether you reject the null hypothesis at a selected significance level of $\alpha = 0.05$.
- 6.4 EASY** Define what a 'critical value' signifies within a standard normal Z-test rejection region topology.
- 6.5 EASY** Explain the true mathematical meaning of a significance level boundary parameter ($\alpha = 0.01$) chosen for an anomaly or fraud detection model validation test.
- 6.6 EASY** Classify the following statement as belonging to a Null (H_0) or Alternative (H_1) framework: 'The new graph neural network architecture does not change the mean customer image processing latency.'
- 6.7 EASY** If an e-commerce feature A/B test returns an aggregate p-value of **0.45**, what can you scientifically conclude about the observed metric difference between your group means?
- 6.8 EASY** A machine learning engineer shifts the alpha significance threshold parameter α from **0.05** down to **0.01**. Describe whether this modification decreases or increases the risk of triggering a Type I error.
- 6.9 EASY** State the universal baseline null hypothesis formulation when verifying if the calculated Pearson product-moment correlation coefficient between two input features is statistically significant.
- 6.10 EASY** Explain what an active test statistic measures conceptually relative to the theoretical sampling distribution generated under the null hypothesis assumptions.
- 6.11 EASY** Why is observed raw percentage improvement alone insufficient to declare a statistically sound winner in a web traffic conversion experiment without applying inferential hypothesis testing?
- 6.12 EASY** Identify the specific branch of statistical error committed if a validation pipeline erroneously rejects a perfectly true null hypothesis (H_0).
- 6.13 EASY** Identify the operational error type committed if a data analyst fails to reject a false null hypothesis when an actual true treatment effect is present in the underlying system population.
- 6.14 EASY** For a standard symmetric two-tailed regular Z-test evaluated with $\alpha = 0.05$, state the exact critical boundary threshold Z-scores.
- 6.15 EASY** Define the technical term 'statistical significance' in plain, direct terms suitable for a non-technical corporate product manager reviewing a machine learning feature release experiment.
- 6.16 INTERMEDIATE** In a spam-filtering classification architecture, the prior spam rate is $P(\text{Spam}) = 0.16$. If a specific keyword appears in 80% of spam emails and 10% of safe emails, apply **Bayes' Theorem**.

- 6.17** **INTERMEDIATE** A sample metrics slice of **67** model iterations yields a mean score of 0.78 with standard error equal to 0.02. Calculate the boundaries of a symmetric **95% Confidence Interval**.
- 6.18** **INTERMEDIATE** Differentiate explicitly between **Type I Error (α)** and **Type II Error (β)** contexts within the operational testing of an autonomous model variant numbered **18**.
- 6.19** **INTERMEDIATE** In a spam-filtering classification architecture, the prior spam rate is **$P(\text{Spam}) = 0.19$** . If a specific keyword appears in 80% of spam emails and 10% of safe emails, apply **Bayes' Theorem**.
- 6.20** **INTERMEDIATE** A sample metrics slice of **70** model iterations yields a mean score of 0.78 with standard error equal to 0.02. Calculate the boundaries of a symmetric **95% Confidence Interval**.
- 6.21** **INTERMEDIATE** Differentiate explicitly between **Type I Error (α)** and **Type II Error (β)** contexts within the operational testing of an autonomous model variant numbered **21**.
- 6.22** **INTERMEDIATE** In a spam-filtering classification architecture, the prior spam rate is **$P(\text{Spam}) = 0.22$** . If a specific keyword appears in 80% of spam emails and 10% of safe emails, apply **Bayes' Theorem**.
- 6.23** **INTERMEDIATE** A sample metrics slice of **73** model iterations yields a mean score of 0.78 with standard error equal to 0.02. Calculate the boundaries of a symmetric **95% Confidence Interval**.
- 6.24** **INTERMEDIATE** Differentiate explicitly between **Type I Error (α)** and **Type II Error (β)** contexts within the operational testing of an autonomous model variant numbered **24**.
- 6.25** **INTERMEDIATE** In a spam-filtering classification architecture, the prior spam rate is **$P(\text{Spam}) = 0.25$** . If a specific keyword appears in 80% of spam emails and 10% of safe emails, apply **Bayes' Theorem**.
- 6.26** **INTERMEDIATE** A sample metrics slice of **76** model iterations yields a mean score of 0.78 with standard error equal to 0.02. Calculate the boundaries of a symmetric **95% Confidence Interval**.
- 6.27** **INTERMEDIATE** Differentiate explicitly between **Type I Error (α)** and **Type II Error (β)** contexts within the operational testing of an autonomous model variant numbered **27**.
- 6.28** **INTERMEDIATE** In a spam-filtering classification architecture, the prior spam rate is **$P(\text{Spam}) = 0.28$** . If a specific keyword appears in 80% of spam emails and 10% of safe emails, apply **Bayes' Theorem**.
- 6.29** **INTERMEDIATE** A sample metrics slice of **79** model iterations yields a mean score of 0.78 with standard error equal to 0.02. Calculate the boundaries of a symmetric **95% Confidence Interval**.
- 6.30** **INTERMEDIATE** Differentiate explicitly between **Type I Error (α)** and **Type II Error (β)** contexts within the operational testing of an autonomous model variant numbered **30**.
- 6.31** **INTERMEDIATE** In a spam-filtering classification architecture, the prior spam rate is **$P(\text{Spam}) = 0.31$** . If a specific keyword appears in 80% of spam emails and 10% of safe emails, apply **Bayes' Theorem**.
- 6.32** **INTERMEDIATE** A sample metrics slice of **82** model iterations yields a mean score of 0.78 with standard error equal to 0.02. Calculate the boundaries of a symmetric **95% Confidence Interval**.
- 6.33** **INTERMEDIATE** Differentiate explicitly between **Type I Error (α)** and **Type II Error (β)** contexts within the operational testing of an autonomous model variant numbered **33**.

- 6.34** **INTERMEDIATE** In a spam-filtering classification architecture, the prior spam rate is $P(\text{Spam}) = 0.34$. If a specific keyword appears in 80% of spam emails and 10% of safe emails, apply *Bayes' Theorem*.
- 6.35** **INTERMEDIATE** A sample metrics slice of **85** model iterations yields a mean score of 0.78 with standard error equal to 0.02. Calculate the boundaries of a symmetric **95% Confidence Interval**.
- 6.36** **DIFFICULT** Execute a conceptual blueprint for a **Chi-Square Goodness-of-Fit Test** with **6** degrees of freedom to determine whether observed user distributions match expected baseline ratios.
- 6.37** **DIFFICULT** Formulate the mathematical breakdown equations for an **ANOVA** model framework across **4** distinct feature classes. Show how total sum of squares variance partitions systematically into Between-Group and Within-Group metrics.
- 6.38** **DIFFICULT** Execute a conceptual blueprint for a **Chi-Square Goodness-of-Fit Test** with **8** degrees of freedom to determine whether observed user distributions match expected baseline ratios.
- 6.39** **DIFFICULT** Formulate the mathematical breakdown equations for an **ANOVA** model framework across **6** distinct feature classes. Show how total sum of squares variance partitions systematically into Between-Group and Within-Group metrics.
- 6.40** **DIFFICULT** Execute a conceptual blueprint for a **Chi-Square Goodness-of-Fit Test** with **10** degrees of freedom to determine whether observed user distributions match expected baseline ratios.
- 6.41** **DIFFICULT** Formulate the mathematical breakdown equations for an **ANOVA** model framework across **8** distinct feature classes. Show how total sum of squares variance partitions systematically into Between-Group and Within-Group metrics.
- 6.42** **DIFFICULT** Execute a conceptual blueprint for a **Chi-Square Goodness-of-Fit Test** with **12** degrees of freedom to determine whether observed user distributions match expected baseline ratios.
- 6.43** **DIFFICULT** Formulate the mathematical breakdown equations for an **ANOVA** model framework across **10** distinct feature classes. Show how total sum of squares variance partitions systematically into Between-Group and Within-Group metrics.
- 6.44** **DIFFICULT** Execute a conceptual blueprint for a **Chi-Square Goodness-of-Fit Test** with **14** degrees of freedom to determine whether observed user distributions match expected baseline ratios.
- 6.45** **DIFFICULT** Formulate the mathematical breakdown equations for an **ANOVA** model framework across **12** distinct feature classes. Show how total sum of squares variance partitions systematically into Between-Group and Within-Group metrics.
- 6.46** **DIFFICULT** Execute a conceptual blueprint for a **Chi-Square Goodness-of-Fit Test** with **16** degrees of freedom to determine whether observed user distributions match expected baseline ratios.
- 6.47** **DIFFICULT** Formulate the mathematical breakdown equations for an **ANOVA** model framework across **14** distinct feature classes. Show how total sum of squares variance partitions systematically into Between-Group and Within-Group metrics.
- 6.48** **DIFFICULT** Execute a conceptual blueprint for a **Chi-Square Goodness-of-Fit Test** with **18** degrees of freedom to determine whether observed user distributions match expected baseline ratios.

- 6.49** **DIFFICULT** Formulate the mathematical breakdown equations for an **ANOVA** model framework across **16** distinct feature classes. Show how total sum of squares variance partitions systematically into Between-Group and Within-Group metrics.
- 6.50** **DIFFICULT** Execute a conceptual blueprint for a **Chi-Square Goodness-of-Fit Test** with **20** degrees of freedom to determine whether observed user distributions match expected baseline ratios.

Module 7: Differential Calculus & Machine Learning Optimization

7.1 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 1x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.2 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 2x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.3 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 3x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.4 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 4x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.5 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 5x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.6 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 6x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.7 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 7x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.8 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 8x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.9 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 9x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.10 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 10x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.11 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 11x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.12 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 12x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.13 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 13x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.14 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 14x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.15 EASY Find the first-order derivative for the standalone feature scaling cost function mapping defined by expression $f(x) = 15x^3 - 5x^2 + 7x - 9$ using standard power rules.

7.16 INTERMEDIATE Find the equation of the tangent line touching the non-linear objective function curve $y = x^2 + 16x$ at specific coordinate point location $x = 2$.

- 7.17** **INTERMEDIATE** Apply the product rule of calculus differentiation to calculate the derivative $d/dx [x^2 \cdot \ln(x)]$ for a regularized loss formula.
- 7.18** **INTERMEDIATE** Derive the rate of change derivative expression for the model baseline activation term with scalar variable parameter input x and constant coefficient scaled to **18**.
- 7.19** **INTERMEDIATE** Find the equation of the tangent line touching the non-linear objective function curve $y = x^2 + 19x$ at specific coordinate point location $x = 2$.
- 7.20** **INTERMEDIATE** Apply the product rule of calculus differentiation to calculate the derivative $d/dx [x^5 \cdot \ln(x)]$ for a regularized loss formula.
- 7.21** **INTERMEDIATE** Derive the rate of change derivative expression for the model baseline activation term with scalar variable parameter input x and constant coefficient scaled to **21**.
- 7.22** **INTERMEDIATE** Find the equation of the tangent line touching the non-linear objective function curve $y = x^2 + 22x$ at specific coordinate point location $x = 2$.
- 7.23** **INTERMEDIATE** Apply the product rule of calculus differentiation to calculate the derivative $d/dx [x^8 \cdot \ln(x)]$ for a regularized loss formula.
- 7.24** **INTERMEDIATE** Derive the rate of change derivative expression for the model baseline activation term with scalar variable parameter input x and constant coefficient scaled to **24**.
- 7.25** **INTERMEDIATE** Find the equation of the tangent line touching the non-linear objective function curve $y = x^2 + 25x$ at specific coordinate point location $x = 2$.
- 7.26** **INTERMEDIATE** Apply the product rule of calculus differentiation to calculate the derivative $d/dx [x^{11} \cdot \ln(x)]$ for a regularized loss formula.
- 7.27** **INTERMEDIATE** Derive the rate of change derivative expression for the model baseline activation term with scalar variable parameter input x and constant coefficient scaled to **27**.
- 7.28** **INTERMEDIATE** Find the equation of the tangent line touching the non-linear objective function curve $y = x^2 + 28x$ at specific coordinate point location $x = 2$.
- 7.29** **INTERMEDIATE** Apply the product rule of calculus differentiation to calculate the derivative $d/dx [x^{14} \cdot \ln(x)]$ for a regularized loss formula.
- 7.30** **INTERMEDIATE** Derive the rate of change derivative expression for the model baseline activation term with scalar variable parameter input x and constant coefficient scaled to **30**.
- 7.31** **INTERMEDIATE** Find the equation of the tangent line touching the non-linear objective function curve $y = x^2 + 31x$ at specific coordinate point location $x = 2$.
- 7.32** **INTERMEDIATE** Apply the product rule of calculus differentiation to calculate the derivative $d/dx [x^{17} \cdot \ln(x)]$ for a regularized loss formula.
- 7.33** **INTERMEDIATE** Derive the rate of change derivative expression for the model baseline activation term with scalar variable parameter input x and constant coefficient scaled to **33**.

- 7.34** **INTERMEDIATE** Find the equation of the tangent line touching the non-linear objective function curve $y = x^2 + 34x$ at specific coordinate point location $x = 2$.
- 7.35** **INTERMEDIATE** Apply the product rule of calculus differentiation to calculate the derivative $d/dx [x^{20} \cdot \ln(x)]$ for a regularized loss formula.
- 7.36** **DIFFICULT** Utilize the mathematical *Chain Rule of Derivatives* to calculate the exact parameter derivative expression for composite loss network function $L(w) = (3w^2 + 1)^4$.
- 7.37** **DIFFICULT** In a simple neural network node using linear parameters, a local loss function profile is defined as $E = (y - wx)^2$ with scaling parameter $x = 2$. Find the exact partial derivative gradient vector path $\partial E / \partial w$ used in updating weights.
- 7.38** **DIFFICULT** Utilize the mathematical *Chain Rule of Derivatives* to calculate the exact parameter derivative expression for composite loss network function $L(w) = (3w^2 + 3)^4$.
- 7.39** **DIFFICULT** In a simple neural network node using linear parameters, a local loss function profile is defined as $E = (y - wx)^2$ with scaling parameter $x = 4$. Find the exact partial derivative gradient vector path $\partial E / \partial w$ used in updating weights.
- 7.40** **DIFFICULT** Utilize the mathematical *Chain Rule of Derivatives* to calculate the exact parameter derivative expression for composite loss network function $L(w) = (3w^2 + 5)^4$.
- 7.41** **DIFFICULT** In a simple neural network node using linear parameters, a local loss function profile is defined as $E = (y - wx)^2$ with scaling parameter $x = 6$. Find the exact partial derivative gradient vector path $\partial E / \partial w$ used in updating weights.
- 7.42** **DIFFICULT** Utilize the mathematical *Chain Rule of Derivatives* to calculate the exact parameter derivative expression for composite loss network function $L(w) = (3w^2 + 7)^4$.
- 7.43** **DIFFICULT** In a simple neural network node using linear parameters, a local loss function profile is defined as $E = (y - wx)^2$ with scaling parameter $x = 8$. Find the exact partial derivative gradient vector path $\partial E / \partial w$ used in updating weights.
- 7.44** **DIFFICULT** Utilize the mathematical *Chain Rule of Derivatives* to calculate the exact parameter derivative expression for composite loss network function $L(w) = (3w^2 + 9)^4$.
- 7.45** **DIFFICULT** In a simple neural network node using linear parameters, a local loss function profile is defined as $E = (y - wx)^2$ with scaling parameter $x = 10$. Find the exact partial derivative gradient vector path $\partial E / \partial w$ used in updating weights.
- 7.46** **DIFFICULT** Utilize the mathematical *Chain Rule of Derivatives* to calculate the exact parameter derivative expression for composite loss network function $L(w) = (3w^2 + 11)^4$.
- 7.47** **DIFFICULT** In a simple neural network node using linear parameters, a local loss function profile is defined as $E = (y - wx)^2$ with scaling parameter $x = 12$. Find the exact partial derivative gradient vector path $\partial E / \partial w$ used in updating weights.
- 7.48** **DIFFICULT** Utilize the mathematical *Chain Rule of Derivatives* to calculate the exact parameter derivative expression for composite loss network function $L(w) = (3w^2 + 13)^4$.

7.49 **DIFFICULT** In a simple neural network node using linear parameters, a local loss function profile is defined as $E = (y - wx)^2$ with scaling parameter $x = 14$. Find the exact partial derivative gradient vector path $\partial E / \partial \mathbf{w}$ used in updating weights.

7.50 **DIFFICULT** Utilize the mathematical *Chain Rule of Derivatives* to calculate the exact parameter derivative expression for composite loss network function $L(\mathbf{w}) = (3w^2 + 15)^4$.